



AMTEC

AGRICULTURAL MACHINERY TESTING AND EVALUATION CENTER

TEST REPORT NO. 2013 – 30



HANDY Shredder
Model: Cutlass 7.0 Hp

HANDY CUTLASS 7.0 HP SHREDDER

Test Report No.	:	2013 - 30
Date of Test	:	February 28, 2013
Test Valid Until	:	February 28, 2018
Brand	:	HANDY
Model	:	Cutlass 7.0 HP
Type	:	Vertical single shaft, wheelbarrow-type
Manufacturer	:	HANDY 4-IN-1 INC. 116 4 th St., Brgy. Mariana New Manila, Quezon City
Test Requested by	:	Manufacturer
Machine Classification	:	Commercial

SUMMARY

The HANDY Cutlass 7.0 Hp Shredder was tested using mixture of chopped and split “buko” husk with shell with an average Moisture Content of 80.7%.

The HANDY Cutlass 7.0 Hp Shredder had an average Input Capacity of 832.8 kg/h of “buko” husks with shell at blade shaft speed of 3080 rpm.

The Fuel Consumption Rate of the 5.22 kW TECUMSEH ENDURO XL gasoline engine was 1.45 L/h.

Three persons were needed to operate the machine.

PURPOSE AND SCOPE OF TEST

The machine was tested for verification of its performance. The test was conducted in Brgy. Mariana, New Manila, Quezon City on February 28, 2013. The test was carried out to assess and verify the following:

1. Verification of machine specifications
2. Performance
 - a. Input capacity
 - b. Speed of components
 - c. Noise emitted by the machine
 - d. Fuel Consumption
3. Labor requirement

OPERATOR'S MANUAL

No Operator's Manual was submitted.

METHOD OF TEST

The shredder was tested following the "PAES 245-2010, Agricultural Machinery – Biomass Shredders – Methods of Test".

DESCRIPTION OF THE MACHINE

The HANDY Cutlass 7.0 Hp as shown in **Figure 1**, was a vertical single shaft, wheelbarrow-type shredder. It was powered by a 5.22 kW TECUMSEH ENDURO XL gasoline engine which drove directly the machine's blade systems through a belt-pulley transmission system. The machine consisted of the hopper and the shredding/chipping chamber.

The hopper (**Fig. 2**) was where the materials to be shred were fed. It was funnel-shaped and was hinged to the shredding section. Material to be shred was fed into the hopper to a system of rotating blades and ejected at the front of the machine.

The shredding section (**Fig. 3**) consisted of one piece wing blade, one piece S-blade and two pieces corrugated blades. This section was also provided with two pieces stationary counter blades.

Below the shredding section was the chipping section (**Fig. 4**) for woods which consisted of a chipper plate and chipper blades. Wood to be chipped was fed to a 43 mm diameter opening (**Fig. 5**) which was provided at the side of the machine.

The machine was a wheelbarrow-type provided with two front rubber wheels (**Fig. 6**) and handles (**Fig. 7**) used during transport.

Detailed specifications of the machine are found in **Table 1**.



Figure 1. HANDY Shredder
Model: Cutlass 7.0 Hp



Figure 2. Hopper

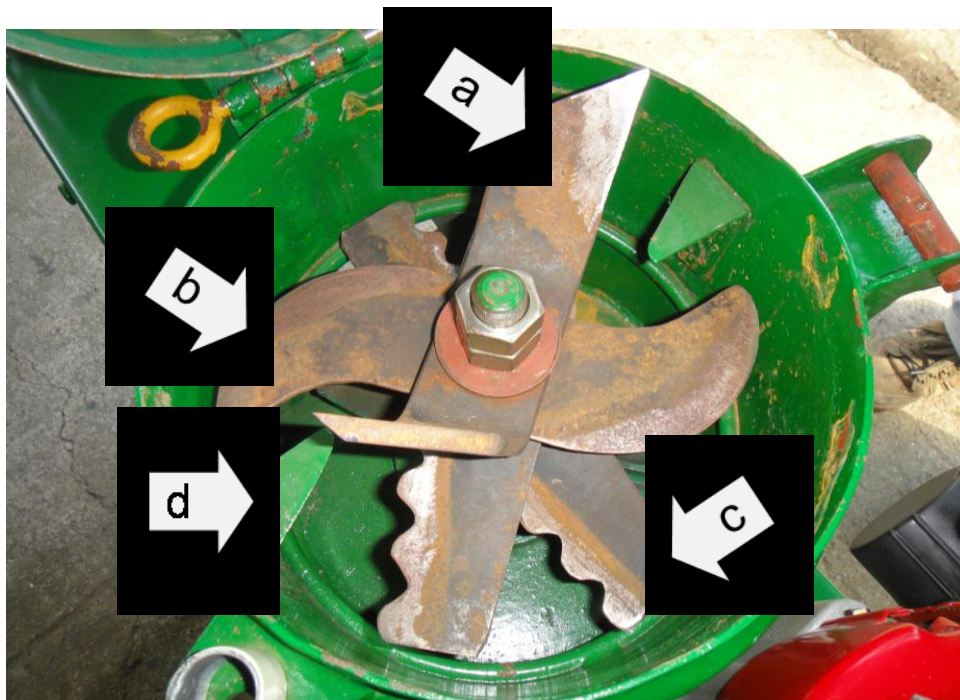


Figure 3. a) Wing blade
b) S-blade
c) Corrugated blade
d) Counter blade

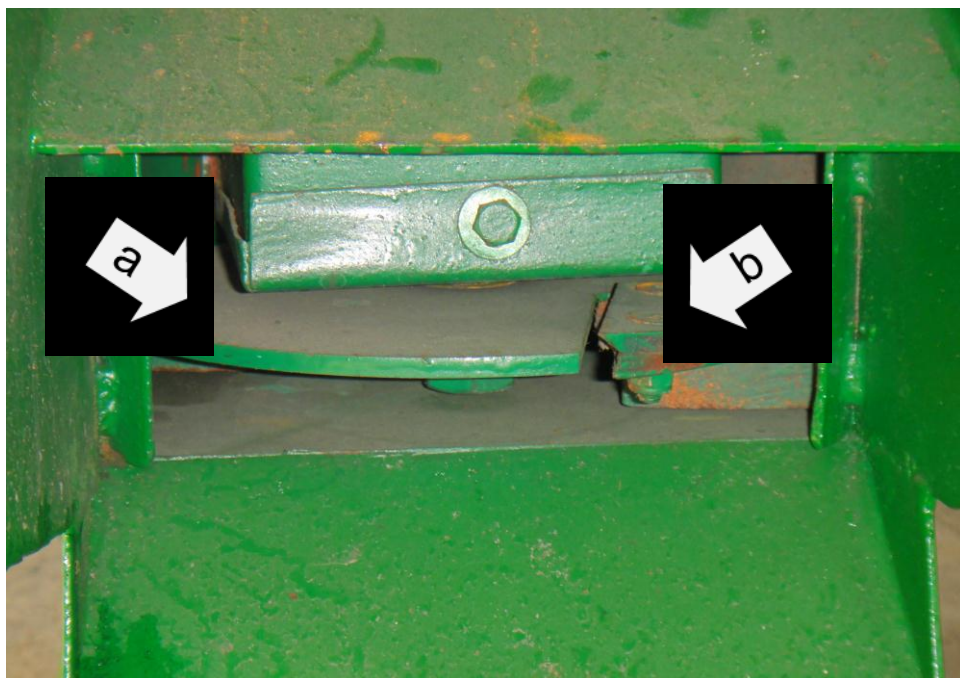


Figure 4. a) Chipping plate
b) Chipping blade

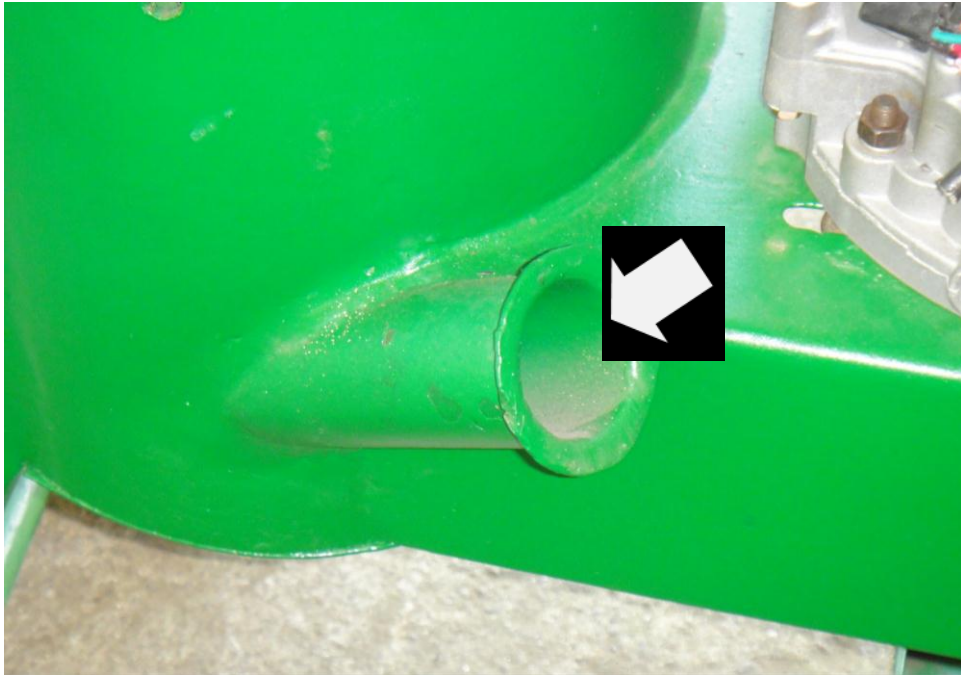


Figure 5. Chipping inlet opening

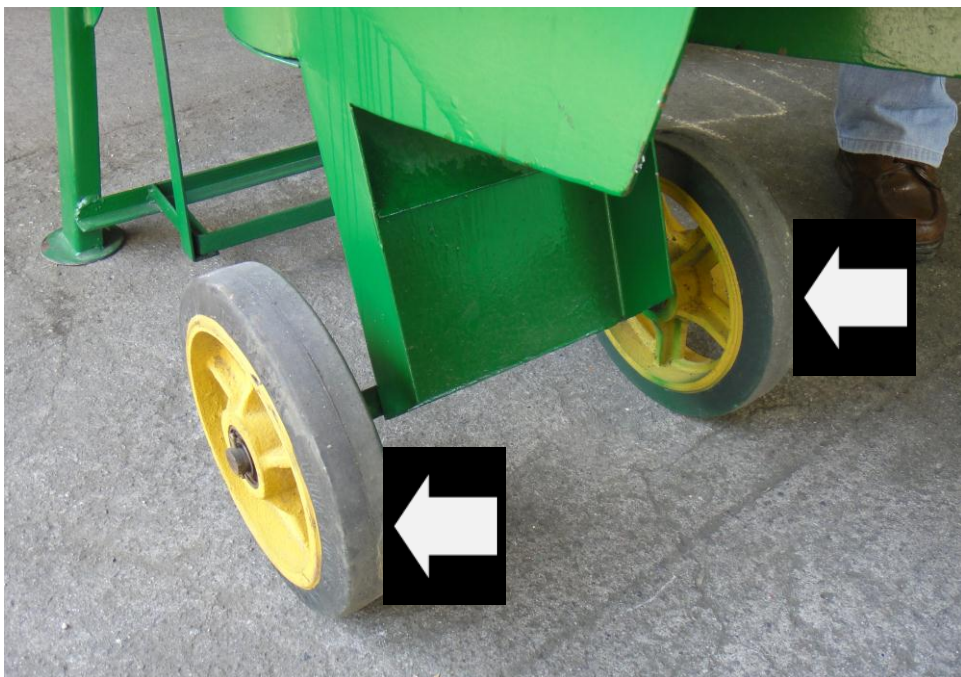


Figure 6. Rubber wheels

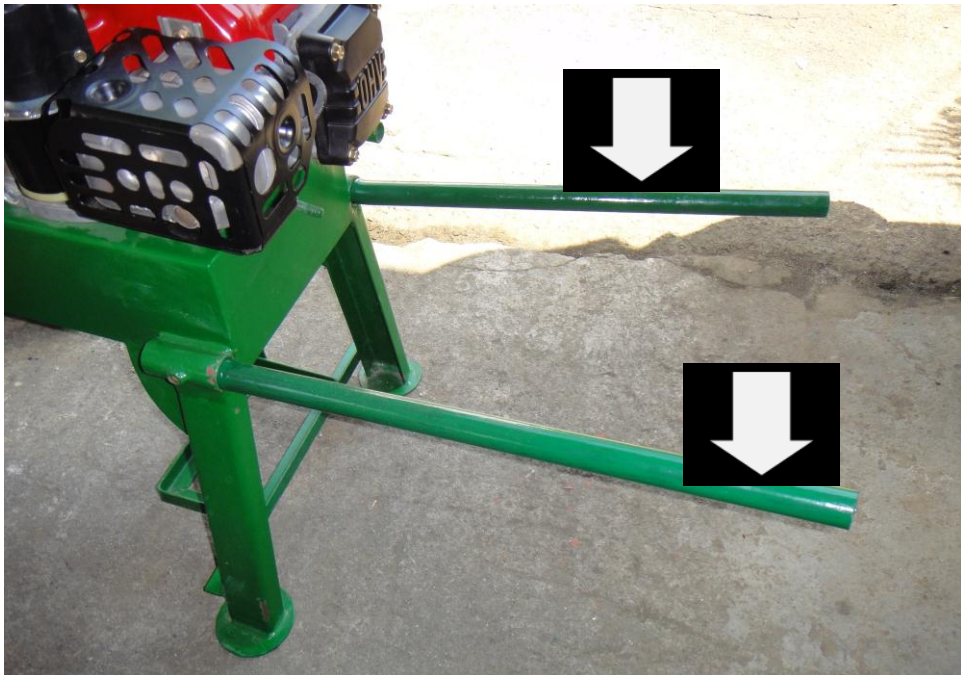


Figure 7. Handles

MACHINE SPECIFICATIONS

Table 1. Detailed specifications of the HANDY Cutlass 7.0 Hp Machine

A.	Brand	:	HANDY
B.	Model	:	Cutlass 7.0 Hp
C.	Serial No.	:	None
D.	Manufacturer	:	HANDY 4-IN-1 INC. 116 4 th St., Brgy. Mariana New Manila, Quezon City
E.	Overall Dimensions and Weight		
1.	Length (mm)	:	890
2.	Width (mm)	:	530
3.	Height (mm)	:	1250
4.	Weight (kg) without engine	:	No Data
F.	Hopper		
1.	Type	:	Funnel shape
2.	Funnel bottom opening (mm)	:	250
3.	Material	:	Cold-Roll Sheet Ga. #16
G.	Shredding cylinder housing		
1.	Size, L x D (mm)	:	195 x 250
2.	Material	:	MS Plate 2.5 mm thick

Table 1. Detailed specifications of the HANDY Cutlass 7.0 Hp (cont'n.)

H.	Corrugated Blade		
1.	No. of pieces	:	2
2.	Size, L x W x T (mm)	:	240 x 60 x 3
3.	Material	:	Tool steel (3 mm thick)
I.	Wing Blade		
1.	No. of pieces	:	1
2.	Material	:	Spring steel (7 mm thick)
J.	S-Blade		
1.	No. of pieces	:	1
2.	Size, L x W x T (mm)	:	240 x 60 x 3
3.	Material	:	Tool steel (3 mm thick)
K.	Counter Blade		
1.	No. of pieces	:	2
2.	Size, L x W x T (mm)	:	50 x 35 x 3
L.	Chipper Blade		
1.	No. of pieces	:	2
2.	Material	:	Bohler steel AISI-01 (6 mm thick)
M.	Chipper Plate		
1.	Material	:	MS Plate (6 mm thick)

Table 1. Detailed specifications of the HANDY Cutlass 7.0 Hp (cont'n.)

N.	Output chute		
	1.	Size of opening, L x W (mm)	: 230 x 155
	2.	Height from the ground (mm)	: 455
O.	Engine Pulley	:	102 mm x 2A x 19 mm
P.	Shredding Shaft Pulley	:	102 mm x 2A x 19 mm
Q.	Size of Belt	:	A - 32
R.	Rubber Wheels		
	1.	No. of pieces	: 2
	2.	Size (D x W), mm	: 250 x 55
S.	Engine (Prime mover)		
	1.	Brand	: TECUMSEH
	2.	Model	: ENDURO XL
	3.	Make	; Canada
+	4.	Rated power (kW)	: 5.22
	5.	Rated speed (rpm)	: 3600
	6.	Type	: Four-stroke, vertical crankshaft, OHV, single cylinder gasoline engine
	7.	Starting system	: Rope recoil
	8.	Cooling system	: Air-cooled

RESULTS AND DISCUSSIONS

Results of performance tests are shown in **Table 2**.

Capacity

Test results showed that the machine had an Input Capacity of 832.8 kg/h of “buko” husks with shell at blade shaft speed of 3080 rpm.

Fuel Consumption

The 5.22 kW TECUMSEH ENDURO XL gasoline engine had a Fuel Consumption Rate of 1.45 L/h.

Labor Requirement

Three persons (**Fig. 8**) were needed to operate the machine.

Observations:

1. Mixture of split and chopped “buko” husks with shell (**Fig. 9**) was used in the test. Totally shredded “buko” husks (**Fig. 10**) were produced.



Figure 8. Operators of the machine



Figure 9. “Buko” husks with shell



Figure 10. Shredded “buko” husks

PERFORMANCE TEST

Machine Tested : HANDY Shredder
Model: Cutlass 7.0 Hp

Date of Test : February 28, 2013

Location of Test : New Manila, Quezon City

Test Material Used : Mixture of split and chopped “buko” husk with shell

Table 2. Conditions of Test Materials

ITEMS	Data
1. Moisture Content, MC _{w.b.} (%)	80.7
2. Dimensions, L x W x T (mm)	149 x 121 x 35

Table 3. Result of Performance Test

ITEMS	Data
1. Speed of Components	
a) Engine (rpm)	
Without load	3210
With load	3134
b) Blade Assembly Shaft	
Without load	3186
With load	3080
2. Noise Level, dB(A)	
Without load	93.1
With load	98.2
3. Prime mover's Fuel Consumption (5.22 kW TECUMSEH ENDURO XL Gasoline Engine), L/h	1.45
4. Input Capacity, kg/h	832.8

Test Engineer:

DARWIN C. ARANGUREN

Checked by:

ROMULO E. EUSEBIO

Approved for Release:

DELFIN C. SUMINISTRADO
Director

Date

N.B.

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